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Combustor Diffuser Modeling
Part II: Inlet Profiles & 3-D Calculations

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ABSTRACT

3-D simulation of a typical gas turbine diffusion system is conducted with the standard k-ε model along with the non-equilibrium wall function of FLUENT® software. The diffuser inlet conditions were estimated by using through-the-rig inlet 3-D calculations including the bow-tie profilers. Calculations were compared with a typical 3-D diffuser rig data for the three inlet profiles, namely center-, inner- and outer-peak velocity profiles. Back-to-back comparison with 2-D calculations clearly showed superior performance achieved with 3-D simulation.

INTRODUCTION

2-D and 3-D CFD calculations are generally done during the gas turbine combustion system design process to aid in the optimization of the diffusion system. Early work by Karki et al[1] as well as the follow-on publications, e.g., Reference 2 have shown that the standard k-ε model with wall functions overpredicts C_P by 5-15% compared to data. A preliminary assessment of various advanced turbulence models indicates[3] that SST k- ω turbulence model has shown potential for giving better agreement with data for flows subject to adverse pressure gradients including separation. However due to lack of systematic validation of CFD-based approach with data of combustor diffusion systems of practical interest to designers, the current design process relies on empirical data collected on simulated full-scale 3-D diffuser rigs coupled with qualitative guidance from CFD.

Prompted by recent advances in CFD simulation technology, turbulence modeling[4,5] and availability of an excellent data set[6], we have undertaken a comprehensive investigation to study the effect of the following on C_P and Ω:

1. Diffusion system design parameters including inlet profiles, pre-diffuser area ratio, L/H, cowling, blockages due to dome swirlers, nozzle stem and tip, airflow splits between the dome, inner and outer annuli of the combustor liner.
2. Procedures for calculating diffuser inlet conditions.
3. 2-D versus 3-D simulations.
4. Different turbulence models and near-wall treatments to capture boundary layer losses and flow separation.

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The Part I paper[6], focuses on the procedures for calculating the inlet profiles specific to Runs 3, 4 and 5 of Table 1 and determine how far one can go with 2-D simulations, the standard k-ε model along with the two near-wall treatments, and SST k-ω turbulence model coded in FLUENT®[5]. Best agreement with data is obtained with the standard k-ε model starting with the calculated diffuser inlet conditions instead of estimating it from 4-5 P_t rakes and the accompanying wall static pressure taps. With measured diffuser inlet profiles and 2-D simulation, the average overprediction for prediffuser CP is 27, 23.5 and 12.5%, respectively for Runs 3, 4 and 5. However, these overpredictions reduce to 7.5, 6 and -2.5%, respectively with the calculated inlet profiles. The corresponding errors with SST k-Ω model are 12.5, 17 and 2.5%. The average underpredictions for Ω are 12.5, 0 are –2.5%. 2-D simulation gave very poor agreement with the dome total pressure recovery coefficient data; prediction is on the average 27.5% lower than data.

This paper, Part II therefore focuses on 3-D calculations for Runs 3, 4 and 5 starting with the 3-D calculated diffuser inlet profiles. The standard k-ε model with non-equilibrium wall function[5] is used for making back-to-back comparison between 2-D reported in Reference 6 and 3-D diffuser simulations presented in this paper.

DATA SETS & INLET CONDITIONS

A typical annular diffuser rig, Figure 1a, consists of a cylindrical annular inlet section, which extended from a rounded entrance to the diffuser inlet. The length of this inlet section from the bell mouth entrance to the prediffuser inlet was ~12.0"or ~16 passage heights. The axial distance from the inlet section entrance to the diffuser inlet measurement plane was ~ 8.7" or ~11 passage heights. The distance from the

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measurement plane to the pre-diffuser inlet was ~3.2" or ~4 passage heights. Within the inlet section at a distance of ~2.5" downstream from the rounded entrance were located the inlet profilers. In the initial rig configuration the profiles consisted of a pair of 360-degree rings designed to protrude 0.10" into the flow stream from both the inner and outer flow path walls, called opposed-ring profiler. The location of these rings placed them ~12 passage heights upstream of the diffuser inlet. The testing reported in this paper used "bow tie" profilers, Fig. 1b. The three shown produce center-, inner- and outer-peak profiles, respectively.

The prediffuser exit angle, exit height, L/H and geometric area are ~10.25^{O}, 1.17", 2.27 and 74.8 in^2, respectively. Cowl opening height, H_{cowl}, geometric area, axial location, L_{cowl} and $(L/H)_{cowl}$ are 0.729", 100 in^2, 2.125" and 2.92, respectively. Outer and inner annuli areas at stations 3.5 (toward the farther end of the annulus) are 39.4 and 30.5 in^2, respectively. The airflow splits between the three streams (dome, inner and outer annuli) are controlled independently in order to evaluate the effect of the flow splits on C_P and Ω.

A total of 118 static and total pressure taps were installed on the diffuser model test rig, shown schematically in Fig. 1a. Of these 64 were located in the rig inlet and prediffuser sections, the remaining 54 were located in the dump regions, the combustor dome, and upstream of the metering orifices. As part of the inlet instrumentation there were four 5-element total pressure rakes along with wall static pressures for each rake. The elements of these rakes were located at y/h = 0.1033, 0.3081, 0.5095, 0.7078, and 0.9031, bottom to top. At the diffuser discharge plane another set of four 5-element total pressure rakes along with associated wall statics were located. The radial positions of these total pressure elements were at y/h = 0.1045, 0.3136, 0.5227, 0.7365, and 0.9233, bottom to top. As part of the combustor dome instrumentation two mixer cups were each instrumented with five total pressures and five static pressures equally spaced around the cup circumferences. These taps were placed to quantify the pressure variation around the mixer cups. Pt3 is based on the calculated value determined from the measured diffuser inlet static pressure and the known inlet mass flow rate, assuming C_D of 1.0. The dome total pressure is taken from the average of the dome pressures mounted onto the dome plate facing upstream, assuming stagnation flow conditions.

For the selected series of 3-D diffuser rig testing to be compared with predictions in this paper and References 7-9, a total of 10 good quality runs (Table 1) are used to quantify the effect of the following on pressure recovery and total pressure loss coefficients, C_P and Ω:

1. Diffuser inlet profiles, Runs 3, 4, 5 and 8 Tables 2 and 3;
2. Diffuser area ratio and length, Runs 3 and 9;
3. W/ and w/o combustor dome cowling, Runs 3 and 6;
4. Cowling inlet area, Runs 3 and 7;
5. Nozzle tip diameter, Runs 3 and 11;
6. Blockage due to nozzle stem, Runs 3 and 12;
7. Airflow splits between dome, inner and outer passages that was conducted for all the 10 test cases.

Here static pressure recovery and total pressure loss coefficients C_P and Ω are:

$$C_P = (P_s - P_{s3})/q_3$$

$$\Omega = (P_{t3} - P_t)/q_3$$

Subscripts s, t and 3 denote static and total pressures, combustor inlet station, respectively; q_3 is the diffuser inlet dynamic head.

Since the measurements were available only at five points across the passage height, reasonable and consistent assumption for the near-wall treatment needs to be made for center, inner and outer peaked profiler runs. The calculated results are therefore strongly influenced by the assumptions made for estimating the inlet conditions.

Gas turbine combustor designers reliance on CFD for pre-test predictions can not be realized without tackling the issues of inlet conditions for the entire combustion system including diffuser inlet, fuel nozzles, mixers, primary/dilution, liner and dome cooling air. Significant progress has been made in this area in the last three years[4]. A similar process has been used to deal with analyzing the bow-tie profiler geometries and attendant 3-dimensional flow field characteristics (Fig. 1b), of each of center, inner and outer peaked profilers were taken up by simulating the test conditions. GAMBIT was used for griding the inlet geometry through the prediffuser. For the through the profiler (TTP) analysis, an all hexahedral mesh comprised of ~512k cells was used, Fig. 1d. A total of 245 grid points were used in the axial direction over a physical length of 9.2" starting from the upstream dump exit to the diffuser inlet. Similarly 20×96 cells were employed at the inlet section (fig. 1c) over a 18^0 sector with a channel height of 0.76". A very fine mesh was employed near the profiler walls with the minimum cell size of 0.0155" and 0.033" in the axial and radial direction respectively. The dimensions of the upstream dump were 8h*5h in the radial and axial direction respectively, where h is the height of the profiler (0.764").

The results reported in the Part I paper[6] are with circumferentially averaged measured and CFD-calculated inlet profiles, refer to Figures 2 and 3 of Reference 6. The Part II paper presents calculations with 2-D inlet profiles to show back-to-back comparison

between 1-D and 2-D inlet profiles as well as 3-D simulation of the diffusion system.

Description of Models & Grids

A commercially available code, FLUENT® was used in our investigation since it already has the turbulence models we had investigated in our earlier studies, e.g., References 1 and 3, in addition to having an excellent griding strategy suitable for the test configuration of interest to us. The reader should refer to FLUENT®'s documentation[5] and the Part I paper[6] for details on the two turbulence models and the two types of wall functions, (standard k-ε and shear stress transport SST k-ω models; standard and non-equilibrium wall functions) used in this study.

Gridding was done in the FLUENT®'s pre-processor, GAMBIT®. An excellent quality all-quad 2d mesh, with 40k cells, was generated for the 2-D axi-symmetric simulation (fig. 1e) ensuring y+ ~40 that is required for using with standard k-ε model with wall-functions. ~10k cells were distributed in the pre-diffuser region, ~6k cells in the inner and outer annulus passages each, and the rest for the dump region. For 3-D simulations, the same 2-D mesh (of ~40k cells) was rotated to give a total of 2.64 million cells to resolve the geometrical details of the fuel nozzle and the concentric simulated mixer as shown in Figure 1e. Some of the critical details of the grids are summarized in the following table.

RESULTS

Table 2 and Figures 2 through 9 show back-to-back comparison between 2-D and 3-D simulations with the standard k-ε model that uses non-equilibrium wall function described in References 5 and 6. For example in Table 2 for Runs 3 and 4 that correspond to the profilers that gave respectively center- and inner-peak velocity profiles at the diffuser inlet. Here the agreement for the prediffuser data is excellent with both 2-D and 3-D simulations, average errors are less than 2.5% for CP, the static pressure recovery coefficient. However, the total pressure loss coefficient Ω is reduce by ~50% with 3-D compared to 2-D modeling approach.

Since it is difficult to simulate 3-D features of fuel nozzles and the concentric mixers properly by 2-D, 2-D modeling does not give good agreement with dome total pressure recovery or loss coefficient being off by as much as 25% for CP and 150% for Ω. On the other hand, the 3-D modeling approach reduces these errors to less than 8% and 50% respectively for CP and Ω. Similarly for both the combustor outer and inner annuli (passages), the average errors by 3-D modeling are ~50% lower than those with 2-D modeling. Moreover, the model's agreement with data in regard to Ω is reasonably well which among other things could be attributed at least partly to the way it has been estimated experimentally.

2.64 M hex+tet grids 1.14 M tets Equiangle skewness< 0.9 Aspect Ratio < 14
Prediffuser 112.5K all hex; 74x29x49 Equiangle skewness< 0.16 y+ 40 approximately
Dump upstream of dome cowling hex+tet grids: 332K 262k hex+70k tets Equiangle skewness< 0.84 y+ 40-120 approximately
Dome cowling region 832K tets Equiangle skewness< 0.9 y+=100-250 approximately
Comb. Outer annulus passage 280K all hex; 97x23x125 cells Equiangle skewness< 0.74 y+= min.1-max.146 approximately, avg. 42
Comb. inner annulus passage hex+tets: 196 K; 95x21-29x65-67 ~ 35k tets Equiangle skewness< 0.83 y+= min. 1- max. 176 approximately, avg. 32
Combustor grids 849K hex cells Equiangle skewness< 0.88 90x77-111, Y-Z cells typical y+=80-150 approximately
Inlet calculations* 2.52M all hex cells Equiangle skewness< 0.9 Aspect ratio< 25 ξ–cells: 114 Upstream of bell mouth 72 Upstream of profiler 238 Downstream of profiler x cells for Leading face of profiler y+= 30-100 approximately
* For SST y+ 5-10 improved by Fluent to achieve y+ around 1

Whereas the modeling does good job for Runs 3 and 4, it does not show higher prediffuser CP and lower Ω values observed experimentally in Run 5, the outer-peak inlet profile. Consequently, the average CP error with the 3-D approach is slightly higher than that with 2-D simulation, being respectively 11.1% and 8.6%. Consequently, the advantage of 3-D over 2-D approach is somewhat less pronounced compared to Runs 3 and 4. Nevertheless, the composite effect for all the three runs as shown in the bottom part of Table 2 still favors 3-D simulation over 2-D modeling approach which is the current design practice for its obvious expediency. Further elaboration is shown in Figure 10 where comparison between data (along x-axis) and model predictions along y-axis for CP and Ω is shown for the prediffuser, dome, outer and inner passages. This figure should be compared with similar Figures 12 and 13 of Reference 6 for the 2-D simulation.

From visual representation, one can conclude the following main points of comparison in Table 3.

1. On an overall basis 3-CFD reproduces data well for Runs 3 and 4, average errors being 1 and 1.5%, respectively, for CP; the range of error being -8 to 10% for Run 3, and -7 to 10% for Run 4. The corresponding errors with 2-D simulation are 7.5%, 6%, (0 to15%) and (-3 to 15%).
2. On the other hand for Run #5, the agreement is not as good; being –7.5% and (0 to –15%) with 3-D, and -2.5 and (5 to –10%) with 2-D simulation.
3. Similar statement can be made for CP of the prediffuser and dome in that the model do better correlation job for Runs 3 and 4 than Run #5. However, for the outer and inner passages the average errors are comparable.
4. 2-D simulation with SST k-ω model is not as good as with the k-ε model, and the both are inferior to that of 3-D simulation.
5. Nevertheless, there is enough disagreement between predictions and CP data from the simulated 3-D diffuser rig, an integrated diffuser and combustor liner analysis for predicting performance, profiles, emissions and wall temperatures should not be expected without applying proper correction at the critical inlets around the combustor to account for deviations between data and model predictions.

Dome Total Pressure Recovery and Ω

Figure 2 shows typical total pressure recovery versus dome flow rate, % compressor flow rate, %Wa3. Data shows that CP decreases from ~0.9 at ~30% flow through the dome to ~0.75 at ~70%Wa3; ~17% decrease in CP. It may be noticed that this paper as well as others in this series of investigation the calculations are also done for several test conditions so that it is possible to draw a trend line to compare with the data. 2-D predictions are considerably lower than data whereas the 3-D simulations for Runs 3 and 4 are quite close to data, achieving qualitative trend rather very well. On the other hand for Run 5, 3-D predictions fall in between 2-D calculations and data. Figure 3 shows comparison between measured and predicted total pressure loss coefficient for the dome. 3-D predictions even though higher than data reproduce experimental trend rather very well.

CP and Ω of Passages

Data for the static pressure recovery and total pressure loss coefficient, CP and Ω was taken near the entrance and discharge planes of both the outer and inner passages, denoted respectively by stations 3.4 and 3.5, see Figure 1a. Figures 4 through 6 show back-to-back comparison between 2-D and 3-D with CP data for the center-, inner- and outer-peak profilers. Clearly 3-D simulations give significantly better agreement with data compared to 2-D approach.

Table summarizes the key findings for the 3-D calculations versus data. Both the experimental and CFD results can be represented well by quadratics with reasonable values of the standard deviation σ, less than 1.5% for the outer passage and 4% for the inner passage recovery coefficients. CFD overpredicts CP by 4 to 10% for Runs 3 and 4 but underpredicts for Run 5 by 5 to 7%. The qualitative behavior for slight increase in CP from the plane 3.4 (near the entrance) to plane 3.5 is captured well by CFD.

	Run 3		Run 4		Run 5	
	Exp	CFD	Exp	CFD	Exp	CFD
Outer Passage Summary						
Correlation σ	1.48%	0.97%	0.89%	0.75%	1.08%	1.26%
Average CP	0.503	0.534	0.510	0.529	0.579	0.536
CFD/Exp		1.061		1.037		0.926
%increase CP 3.4 to 3.5	1.54	1.88	1.57	1.53	2.01	1.77
Inner Passage Summary						
Correlation σ	3.58%	0.72%	3.55%	1.35%	3.38%	0.89%
Average CP	0.521	0.571	0.541	0.588	0.605	0.577
CFD/Exp		1.095		1.085		0.953
%increase CP 3.4 to 3.5	2.52	4.95	2.72	5.37	2.16	4.61

Figures 7, 8 and 9 show measured total pressure loss coefficient versus flow splits for the outer and inner passages at stations 3.4 and 3.5 for the center-, inner- and outer-peak profilers, in addition to the corresponding predictions with 2-D and 3-D modeling techniques. Again 3-D simulations show significantly better performance than the 2-D. From summary given in the following table, we can see that similar to results presented about CP,

- Correlation with quadratics is slightly inferior to that for CP, but is good nevertheless.
- Model underpredicts outer passage Ω by ~5% for Runs 3 and 4 but overpredicts Run 5 by

11%. The corresponding values are slightly higher for the inner passage, being ~10% and 14%.

- The predictions give qualitative good agreement in regard to slight decrease in the loss coefficient at station 3.5 compared to station 3.4.

	Run 3		Run 4		Run 5	
	Exp	CFD	Exp	CFD	Exp	CFD
Outer Passage Summary						
Correlation σ	1.52%	1.15%	0.94%	0.85%	1.48%	1.50%
Average Ω	0.479	0.450	0.471	0.455	0.406	0.451
CFD/Exp		0.940		0.966		1.110
%increase Ω 3.4 to 3.5	-1.59	-2.18	-1.67	-1.80	-2.79	-2.06
Inner Passage Summary						
Correlation σ	4.73%	1.24%	5.11%	1.88%	6.43%	1.38%
Average Ω	0.370	0.333	0.350	0.312	0.294	0.334
CFD/Exp		0.901		0.893		1.136
%increase Ω 3.4 to 3.5	-3.41	-7.91	-4.03	-8.83	-4.26	-7.45

Figure 10 gives visual summary of the data and 3-D predictions by plotting CP and Ω data of the prediffuser, dome, outer and inner passages versus the corresponding predicted values. It clearly shows that predictions though reasonable are still not good enough for undertaking a "black box" integrated diffuser combustor calculations for pretest predictions. However once the design guidelines have been established based on correlation with existing data, it would be possible to estimate whether a particular design would be predicted to have lower or higher performance compared to data, 3-D CFD-based modeling approach would provide the details not easily available through even detail testing effort, as illustrated in the following paragraphs.

CFD PROVIDES INSIGHT

Even with less than majority consensus on the usefulness of 3-D CFD model-based diffusion system design approach, one can't overestimate the importance of having the ability to characterize 3-D flow field, as shown in Figures 11, 12 and 13. When this information suitably combined with a most recent combustion modeling approach[4], it will be possible to explain the occurrence of isolated liner and dome hot spots in addition to quantifying the effect of fuel nozzle manufacturing tolerances on predicting pattern factor.

Finally, comparing 2-D diffusion modeling based predicted distribution of Pt, Ps, CP and turbulence kinetic energy contours, Figure 14 are significantly different from 3-D predictions, Figures 11 and 12. 3-D diffusion system modeling therefore strongly recommended over 2-D diffuser modeling.

SUMMARY

A systematic evaluation of 3-D modeling approach including the diffuser inlet conditions for a given profiler has been conducted to assess its limitations as a design tool for future diffuser design activities. Experimental data from a typical 3-D diffuser rig with three different profilers was compared with both 2-D and 3-D modeling approaches. In both the cases, the diffuser inlet conditions were calculated by doing realistic 3-D simulation of the experimental set-up. Based on earlier 2-D work[6], the standard k-ε model with nonequilibrium wall function was used. Average errors in predicting static pressure recovery coefficient CP for prediffuser (PD), dome (D), outer passage (OP) and inner passage (IP) for the three runs Runs 3 (center-peak), 4 (inner-peak) and 5 (outer-peak) are given in the table:

	Center-Peak	Inner-Peak	Outer-Peak
Prediffuser	2.2%	1.1%	11.1%
Dome	7.7%	6.6%	15.9%
Outer Passage	7.0%	7.1%	7.8%
Inner Passage	10.8%	9.5%	5.0%

Model overpredicts passage recoveries for Runs 3 and 4 by ~9% and underpredicts for Run 5 by ~5%.

Back to back comparison between 3-D and 2-D simulations is summarized in Table 3.

Table 3: Back to back comparison between 3-D and 2-D simulations.

Average, minimum and maximum errors w/ 3-D k-ε model & calculated inlet profiles

		Overall agreement for		% Error in CP of			
		Ω	CP	Prediff	Dome	Outer Passage	Inner Passage
Run #3 (Center-Peak)	Avg.	10.0	1.0	2.4	-7.0	6.8	10.0
	Min	-10.0	-8.0	1.5	-5.0	2.0	6.0
	Max	30.0	10.0	3.3	-9.0	11.5	14.0
Run #4 (Inner-Peak)	Avg.	5.0	1.5	1.5	-4.7	3.0	8.5
	Min	-10.0	-7.0	-7.0	-1.5	0.0	5.0
	Max	20.0	10.0	10.0	-8.0	6.0	12.0
Run #5 (Outer-Peak)	Avg.	65.0	-7.5	-10.5	-15.5	-6.2	-7.5
	Min	10.0	0.0	-9.0	-14.0	-4.0	0.0
	Max	120.0	-15.0	-12.0	-17.0	-8.5	-7.0

Average, minimum and maximum errors w/ 2-D k-ε & SST models & calculated inlet profiles, k-ε model w/ inlet velocity from measured Pt/Ps

		k-ε*		SST*	k-ε**
		Ω	CP	CP	CP
Run #3 (Center-Peak)	Avg.	-12.5	7.5	12.5	27.5
	Min	0.0	0.0	0.0	20.0
	Max	-25.0	15.0	25.0	35.0
Run #4 (Inner-Peak)	Avg.	0.0	6.0	17.0	23.5
	Min	-20.0	-3.0	4.0	15.0
	Max	20.0	15.0	30.0	32.0
Run #5 (Outer-Peak)	Avg.	2.5	-2.5	2.5	12.5
	Min	-10.0	5.0	-10.0	5.0
	Max	15.0	-10.0	15.0	20.0

Error <0.0 means model underpredicts, >0.0 means overpredicts.
* With calculated inlet profiles
**Velocity profiles measured Pt and Ps

From this table and Figures 11 through 14, it is quite clear that 3-D simulation with through-the-rig set-up calculations for estimating the diffuser inlet conditions should be preferred over the current practice: mostly 2-D simulations with inlet velocity profiles estimated from 4-5 total pressure rakes and the accompanying wall static pressure data.

Future Work

Effect of inlet conditions on diffuser separation, and performance of shorter and higher area ratio diffuser is covered in Reference 7, followed by effect of cowling geometry, mixer size and nozzle blockage in Reference 8. Validation of model validation with multi-passage diffusers is presented in References 9 and 10 along with more topics of interest in diffuser design are planned for future work[11-14].

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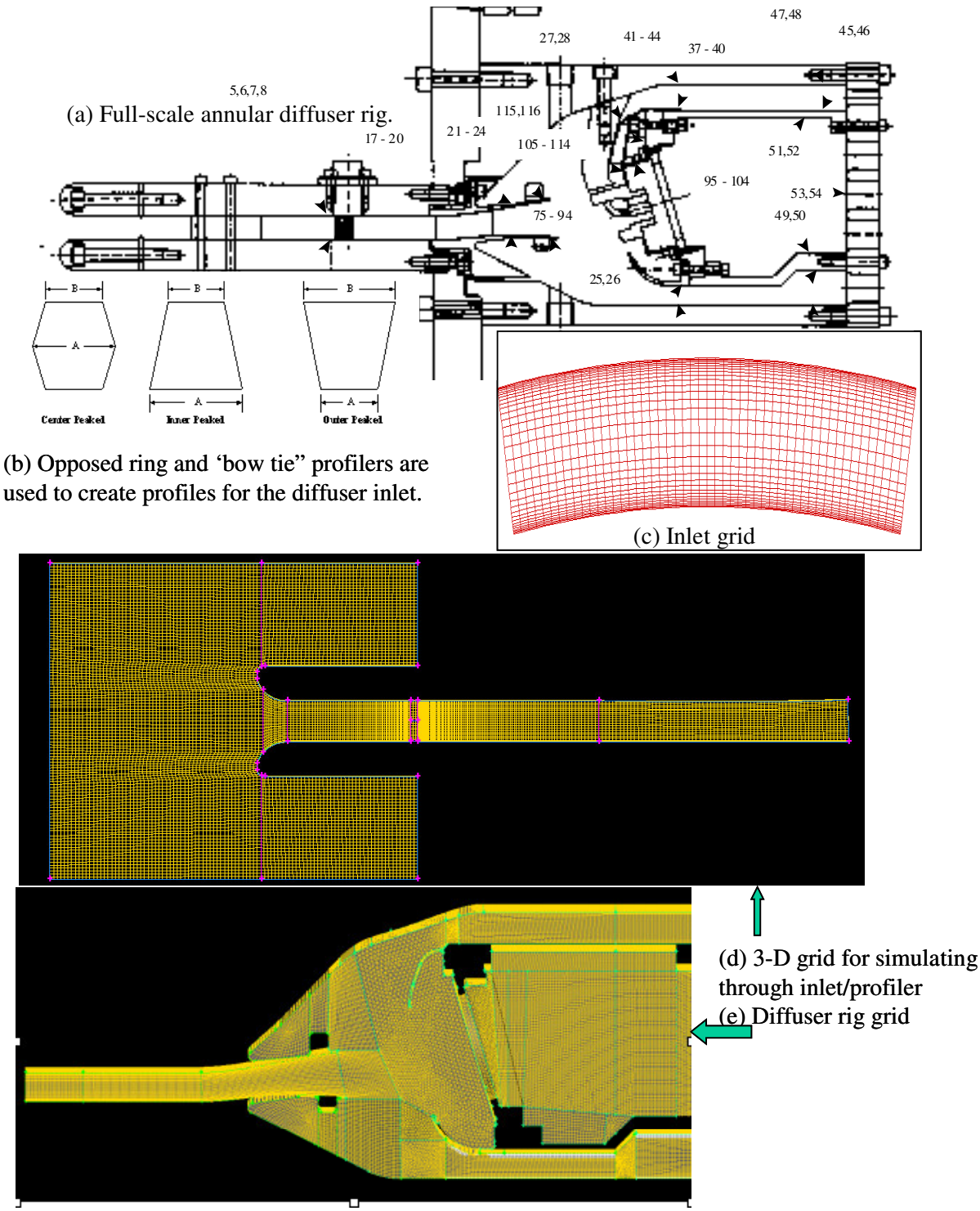
Table 1: Typical 3-D Combustor Diffuser Runs

Run #	Comments
3	Center-peaked inlet profile.
4	Inner-peaked inlet profile.
5	Outer-peaked inlet profile.
10	Repeat of Run #3
6	Center-peaked inlet profile w/o cowling
7	Center-peaked inlet profile w/ reduced cowling open area
8	Opposed ring inlet profiler to compare w/ Run 3
9	Center-peaked profile; increased area shorter diffuser
11	Center-peaked profile; increased nozzle dia to 1.9", slightly modified cowling.
12	Run 11 w/ 20% increased nozzle blockage.

Table 2: Comparison between average data and 2-D & 3-D predictions for C_P and Ω, total pressure loss coefficient at prediffuser exit, dome, outer and inner passages .

	Prediffuser exit		Dome		Outer Passage		Inner Passage	
	CP	Ω	CP	Ω	CP	Ω	CP	Ω
Run 3 Data	0.496	0.138	0.846	0.154	0.505	0.476	0.532	0.361
2-D Pred	0.497	0.123	0.636	0.364	0.575	0.406	0.587	0.308
3-D Pred	0.506	0.127	0.779	0.223	0.539	0.445	0.584	0.319
2-D σ	1.37%	11.53%	25.07%	151.87%	14.50%	15.08%	10.57%	15.57%
3-D σ	2.21%	8.44%	7.74%	50.22%	6.97%	6.64%	10.75%	11.76%
Run 4 Data	0.504	0.129	0.848	0.152	0.516	0.466	0.544	0.343
2-D Pred	0.494	0.157	0.622	0.355	0.562	0.421	0.587	0.302
3-D Pred	0.503	0.123	0.796	0.204	0.533	0.450	0.603	0.298
2-D σ	2.00%	18.87%	21.62%	139.88%	7.14%	7.65%	11.53%	11.64%
3-D σ	1.09%	8.69%	6.56%	47.09%	3.77%	3.64%	9.51%	12.31%
Run 5 Data	0.567	0.068	0.930	0.070	0.582	0.402	0.594	0.289
2-D Pred	0.506	0.097	0.622	0.364	0.604	0.381	0.573	0.330
3-D Pred	0.505	0.127	0.772	0.235	0.541	0.446	0.590	0.321
2-D σ	8.62%	40.44%	24.94%	484.87%	3.80%	4.89%	6.82%	15.86%
3-D σ	11.14%	98.69%	15.86%	321.11%	7.84%	12.07%	5.00%	14.98%
2-D σ	**5.17%**	26.61%	**23.93%**	304.26%	**9.58%**	10.16%	**9.85%**	14.49%
3-D σ	**6.59%**	57.41%	**10.87%**	189.61%	**6.44%**	8.23%	**8.78%**	13.09%

Overall σ for Predictions		
	CP	Ω
2D	**14.04%**	152.97%
3D	**8.37%**	99.35%



(a) Full-scale annular diffuser rig.

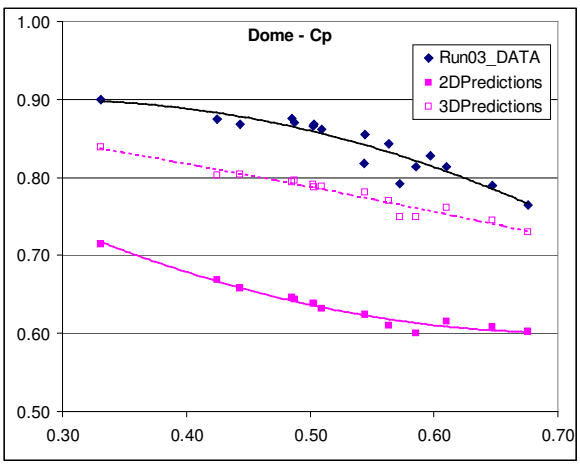
(b) Opposed ring and 'bow tie" profilers are used to create profiles for the diffuser inlet.

(c) Inlet grid

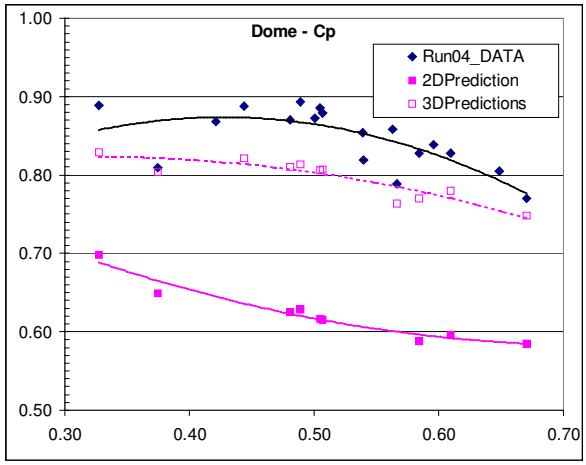
(d) 3-D grid for simulating through inlet/profiler

(e) Diffuser rig grid

Figure 1: Diffuser rig schematic and typical grid used for inlet and diffuser calculations.



(a) Run 3 Center-Peak Velocity Profile.



(b) Run 4 Inner-Peak Velocity Profile.

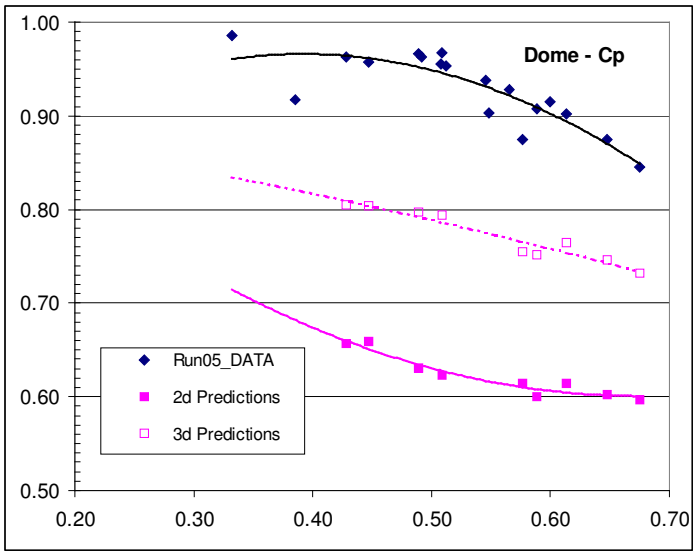
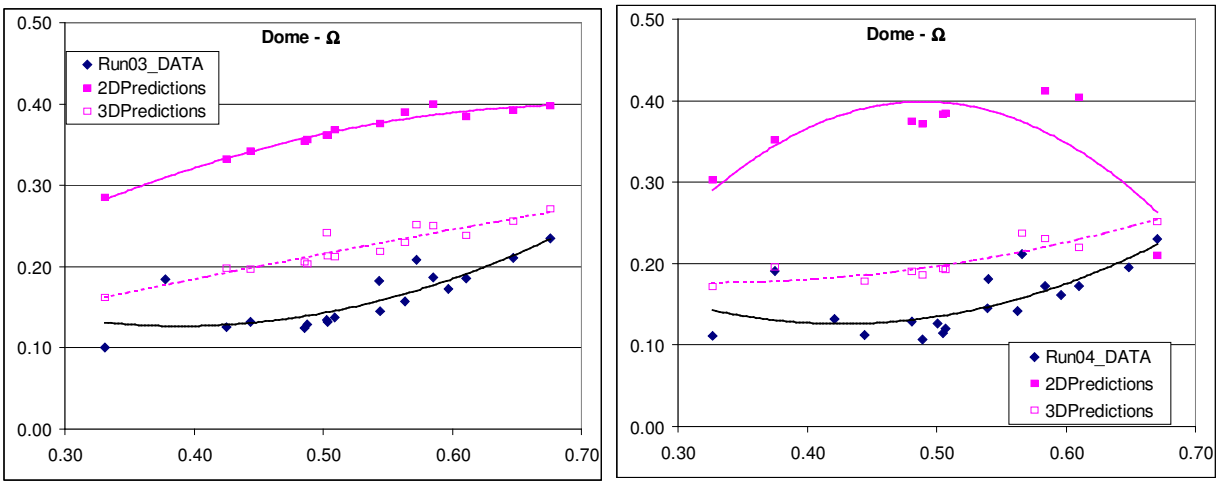
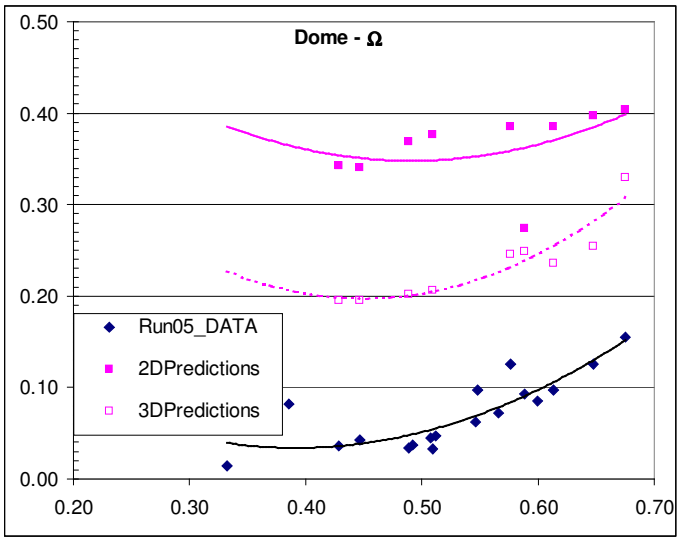


Figure 2: Comparison between data and 2-D and 3-D predictions for dome CP (y-axis) versus flow split through the dome (x-axis).



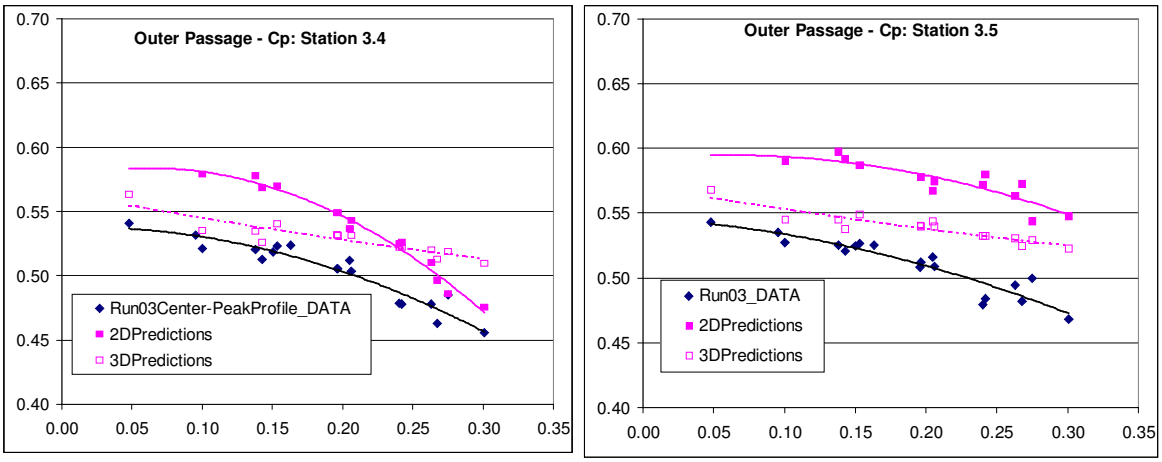
(a) Run 3 Center-peak profile

(b) Run 4 Inner-peak profile

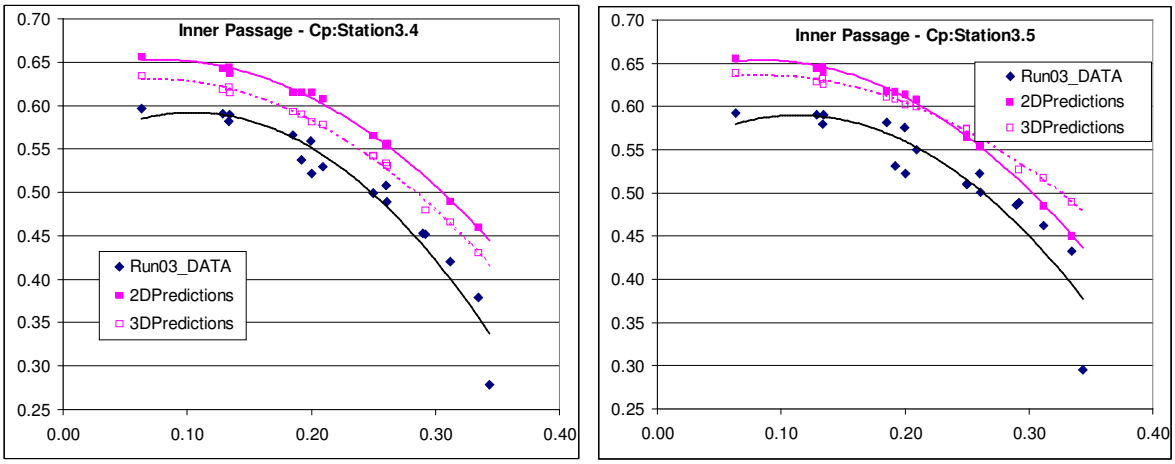


(c) Run5 Outer-Peak Velocity Profile.

Figure 3: Comparison between data and 2-D and 3-D predictions for dome Ω (y-axis), total pressure loss coefficient versus flow split through the dome (x-axis).

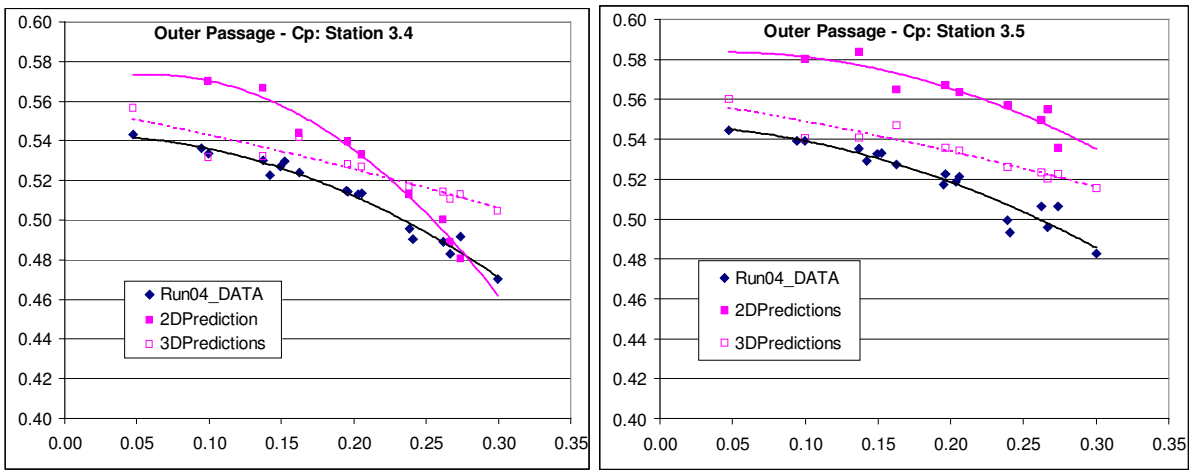


(a) Outer Passage Stations 3.4 and 3.5

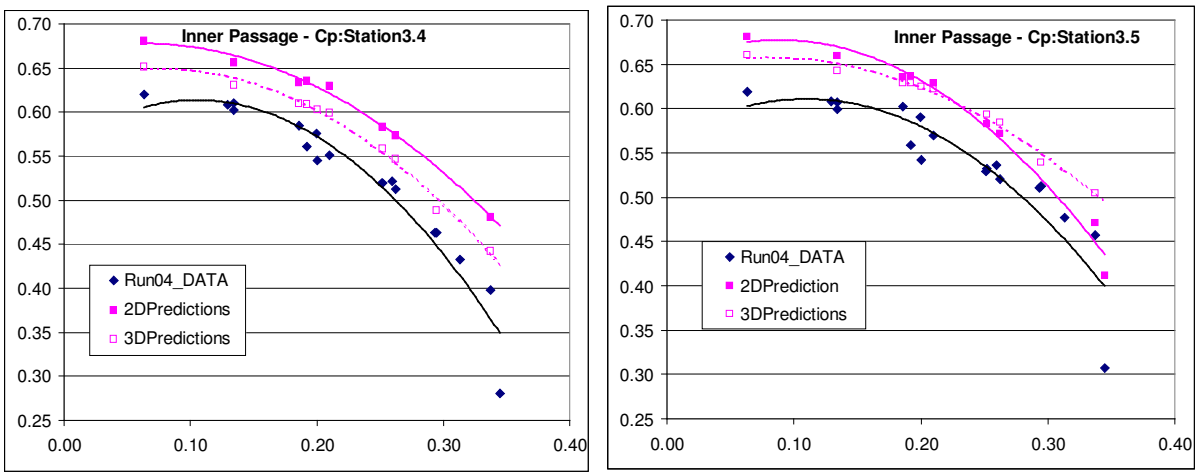


(b) Inner Passage Stations 3.4 and 3.5

Figure 4: Comparison between Run 3 (center-peak inlet profile)data and 2-D and 3-D predictions for CP (y-axis) of combustor outer and inner passages versus flow split for the respective passages (x-axis). Station 3.4 is closer to passage inlet, Station 3.5 toward the end.

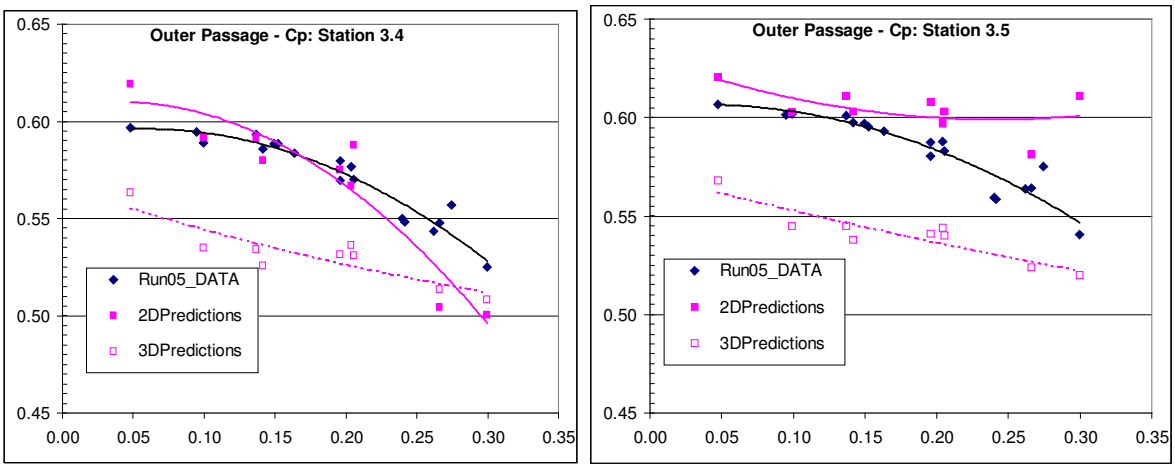


(a) Outer Passage Stations 3.4 and 3.5

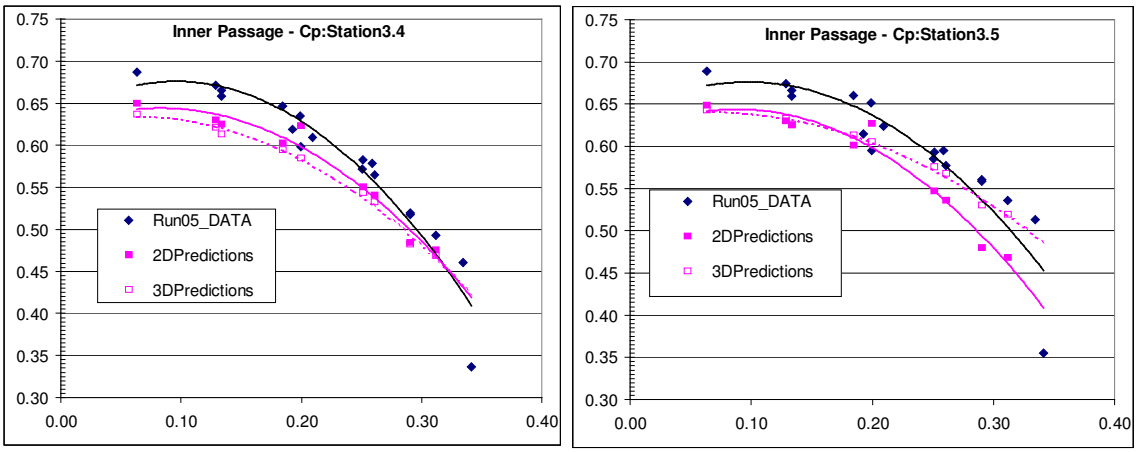


(b) Inner Passage Stations 3.4 and 3.5

Figure 5: Comparison between Run 4 (inner-peak inlet profile) data and 2-D and 3-D predictions for CP (y-axis) of combustor outer and inner passages versus flow split for the respective passages (x-axis). Station 3.4 is closer to passage inlet, Station 3.5 toward the end.

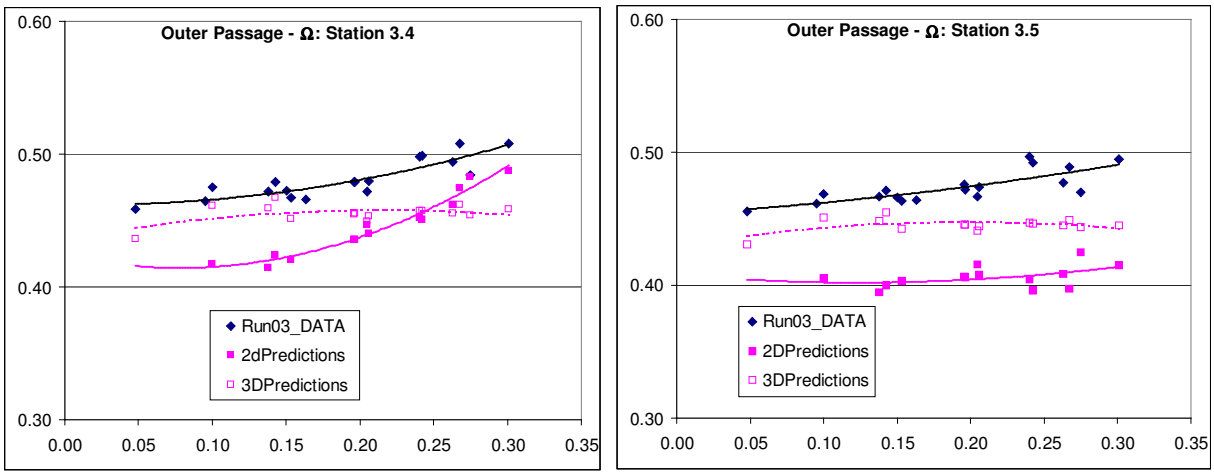


(a) Outer Passage Stations 3.4 and 3.5

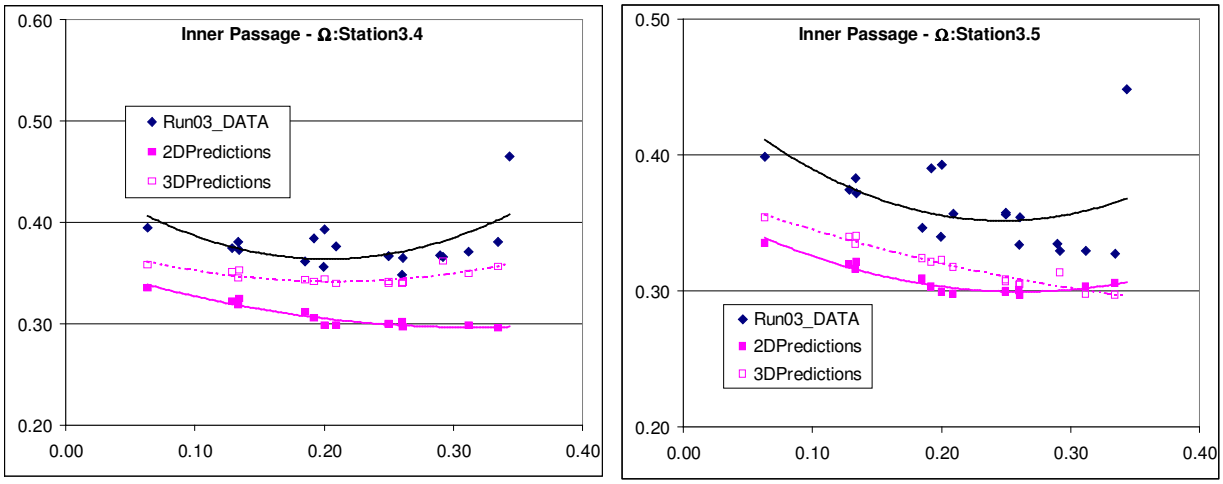


(b) Inner Passage Stations 3.4 and 3.5

Figure 6: Comparison between Run 5 (outer-peak inlet profile)data and 2-D and 3-D predictions for CP (y-axis) of combustor outer and inner passages versus flow split for the respective passages (x-axis). Station 3.4 is closer to passage inlet, Station 3.5 toward the end.

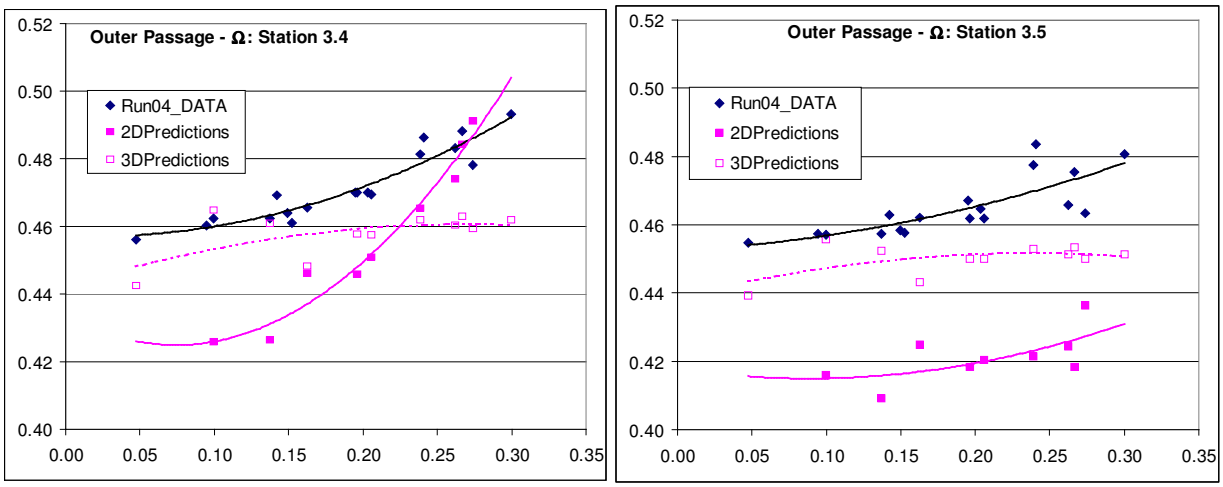


(a) Outer Passage Stations 3.4 and 3.5

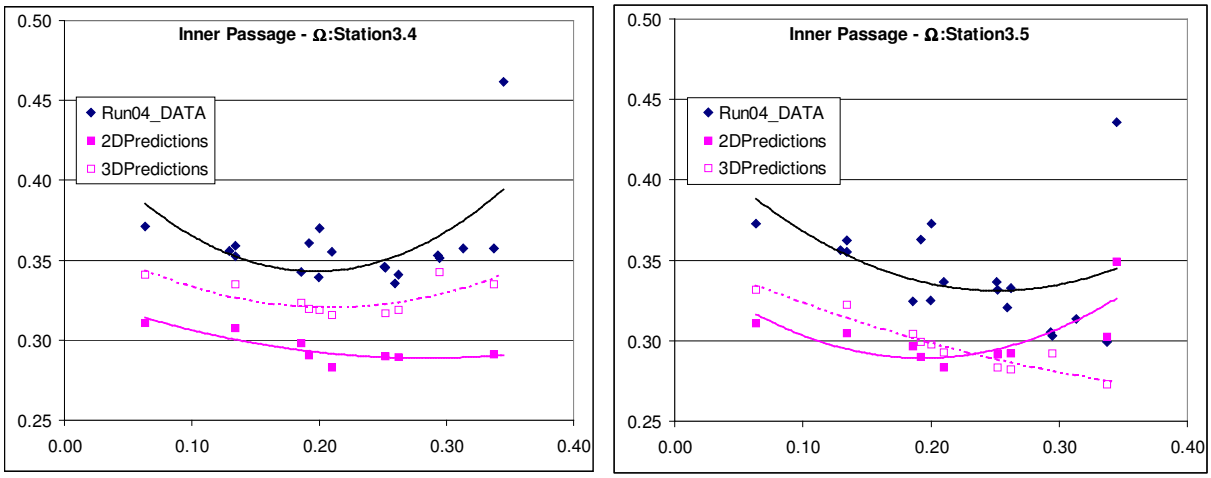


(b) Inner Passage Stations 3.4 and 3.5

Figure 7: Comparison between Run 3 (center-peak inlet profile)data and 2-D and 3-D predictions for Ω (y-axis), total pressure loss coefficient of combustor outer and inner passages versus flow split for the respective passages (x-axis). Station 3.4 is closer to passage inlet, Station 3.5 toward the end.

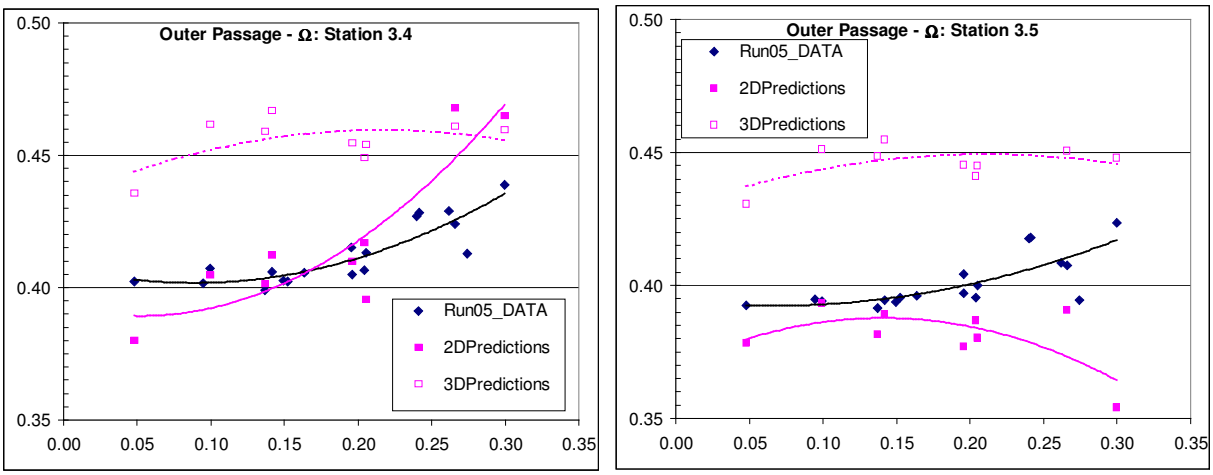


(a) Outer Passage Stations 3.4 and 3.5

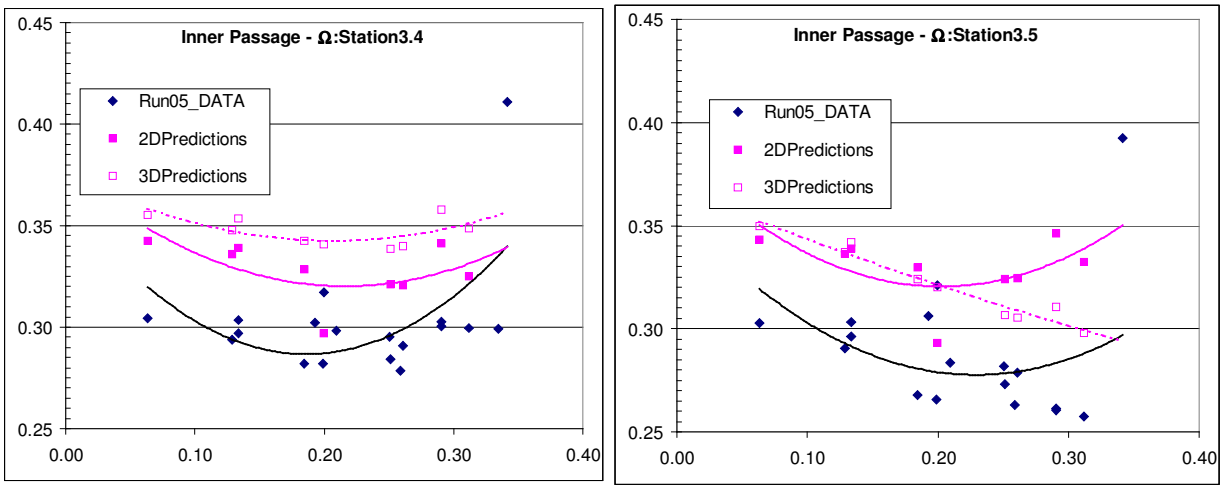


(b) Inner Passage Stations 3.4 and 3.5

Figure 8: Comparison between Run 4 (inner-peak inlet profile)data and 2-D and 3-D predictions for Ω (y-axis), total pressure loss coefficient of combustor outer and inner passages versus flow split for the respective passages (x-axis). Station 3.4 is closer to passage inlet, Station 3.5 toward the end.



(a) Outer Passage Stations 3.4 and 3.5



(b) Inner Passage Stations 3.4 and 3.5

Figure 9: Comparison between Run 5 (outer-peak inlet profile)data and 2-D and 3-D predictions for Ω (y-axis), total pressure loss coefficient of combustor outer and inner passages versus flow split for the respective passages (x-axis). Station 3.4 is closer to passage inlet, Station 3.5 toward the end.

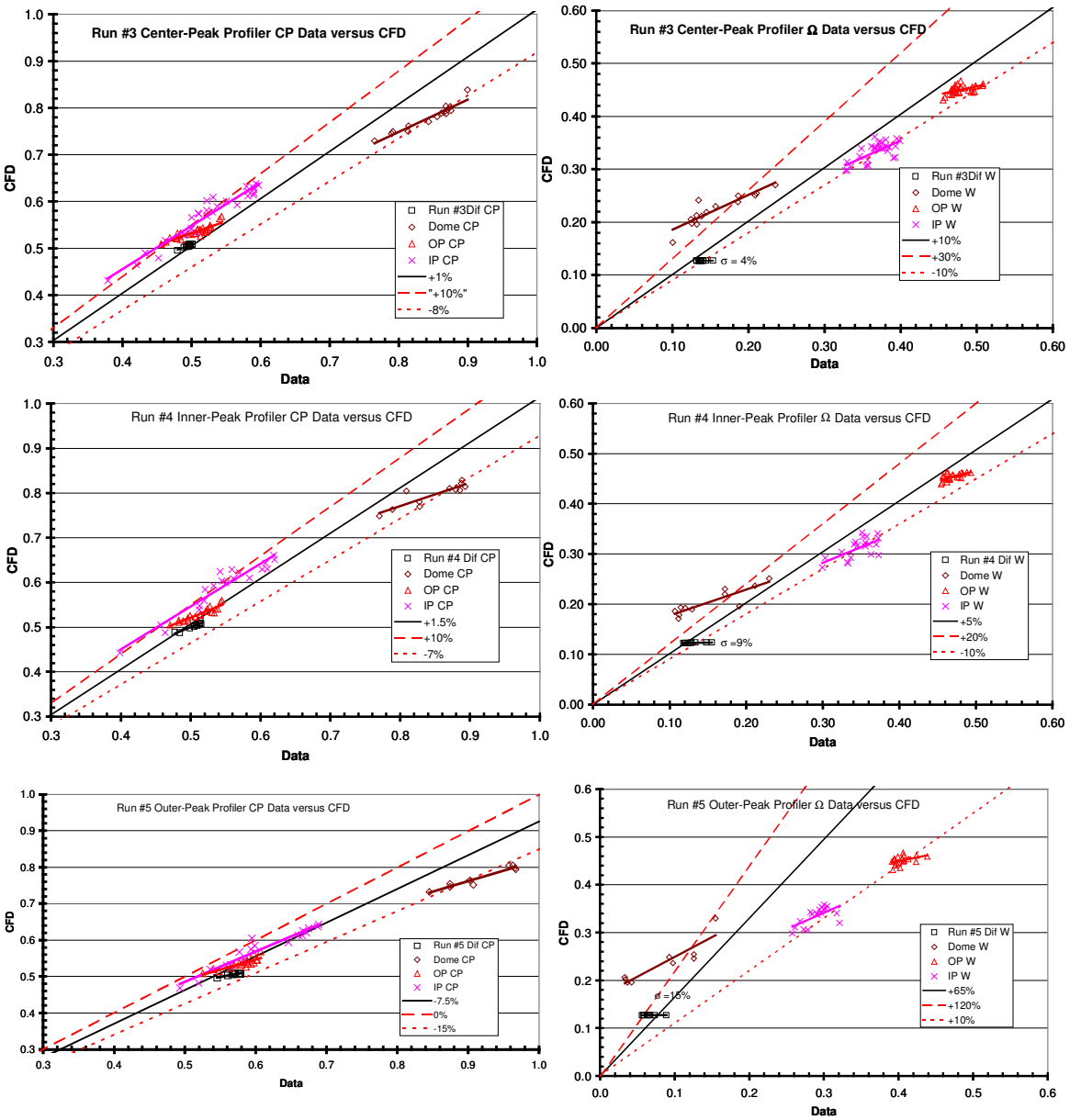


Figure 10: Data versus 3-D CFD prediction for Runs 3, 4 and 5 for CP and total pressure loss Coefficient for the prediffuser, dome, outer and inner passages.

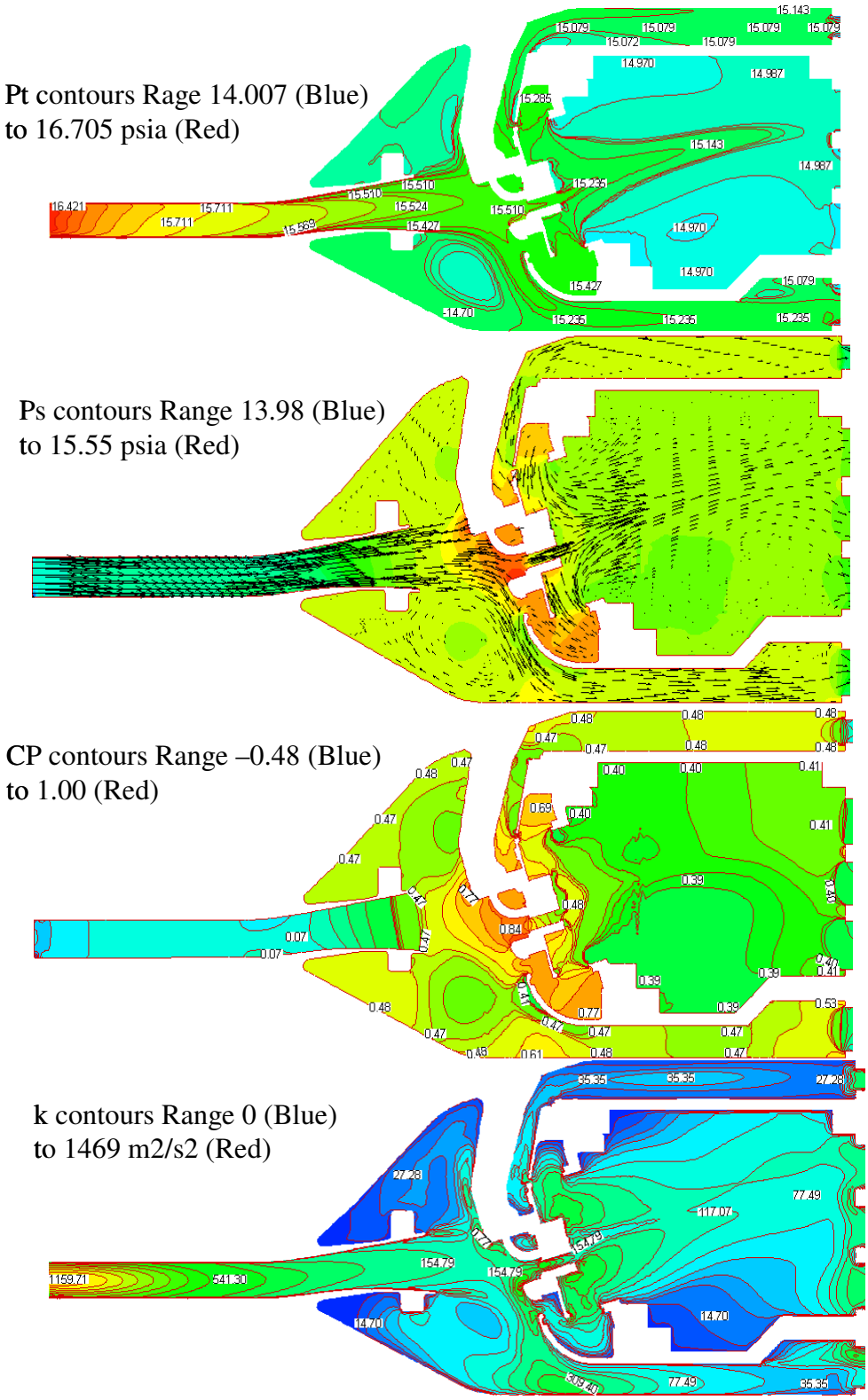


Figure 11: Typical contours of Pt, Ps (superimposed velocity vectors), CP
And turbulence kinetic energy in-line with fuel nozzle stem.

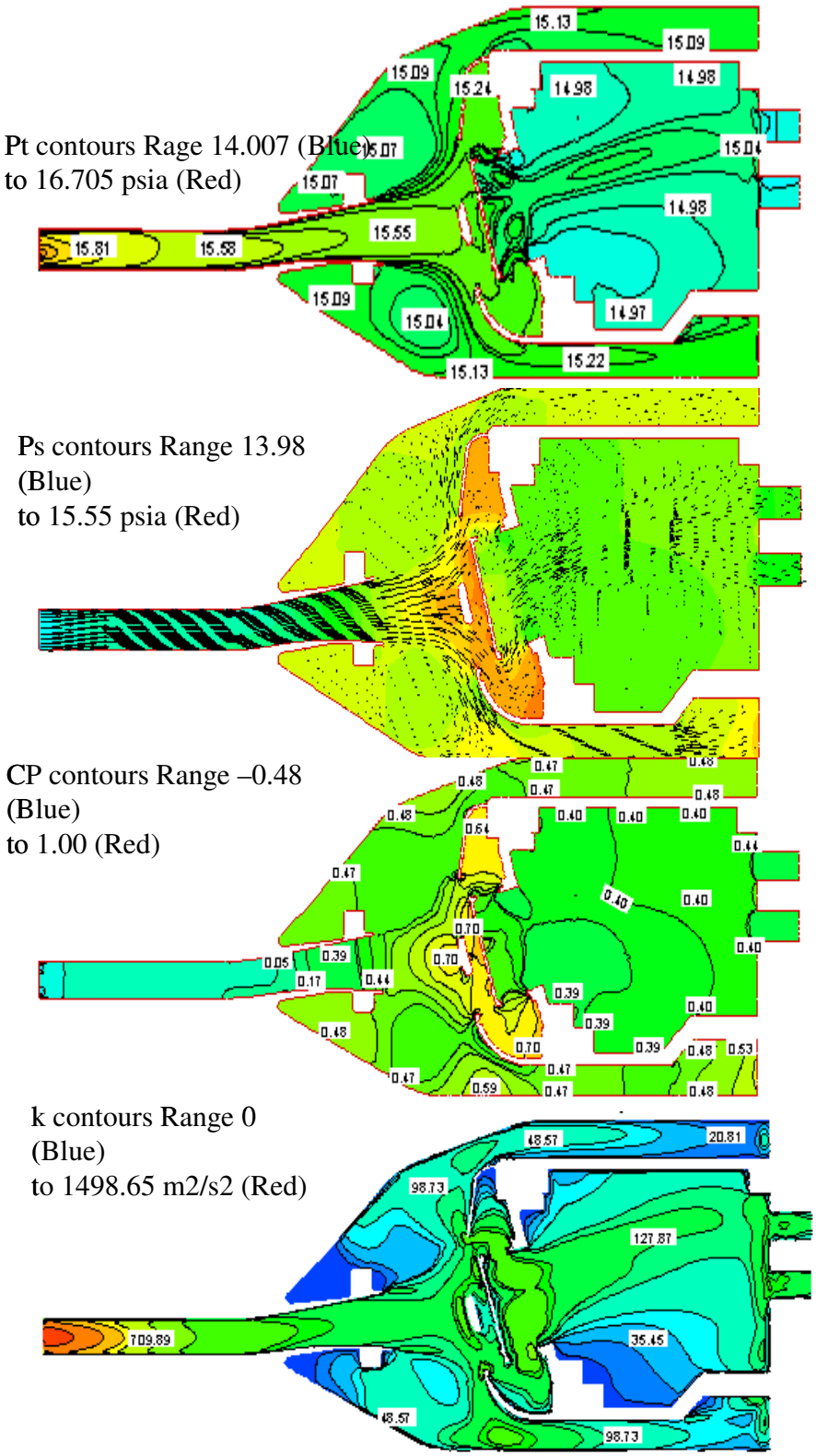


Figure 12: Typical contours of Pt, Ps (superimposed velocity vectors), CP and turbulence kinetic energy in-between fuel nozzle stem.

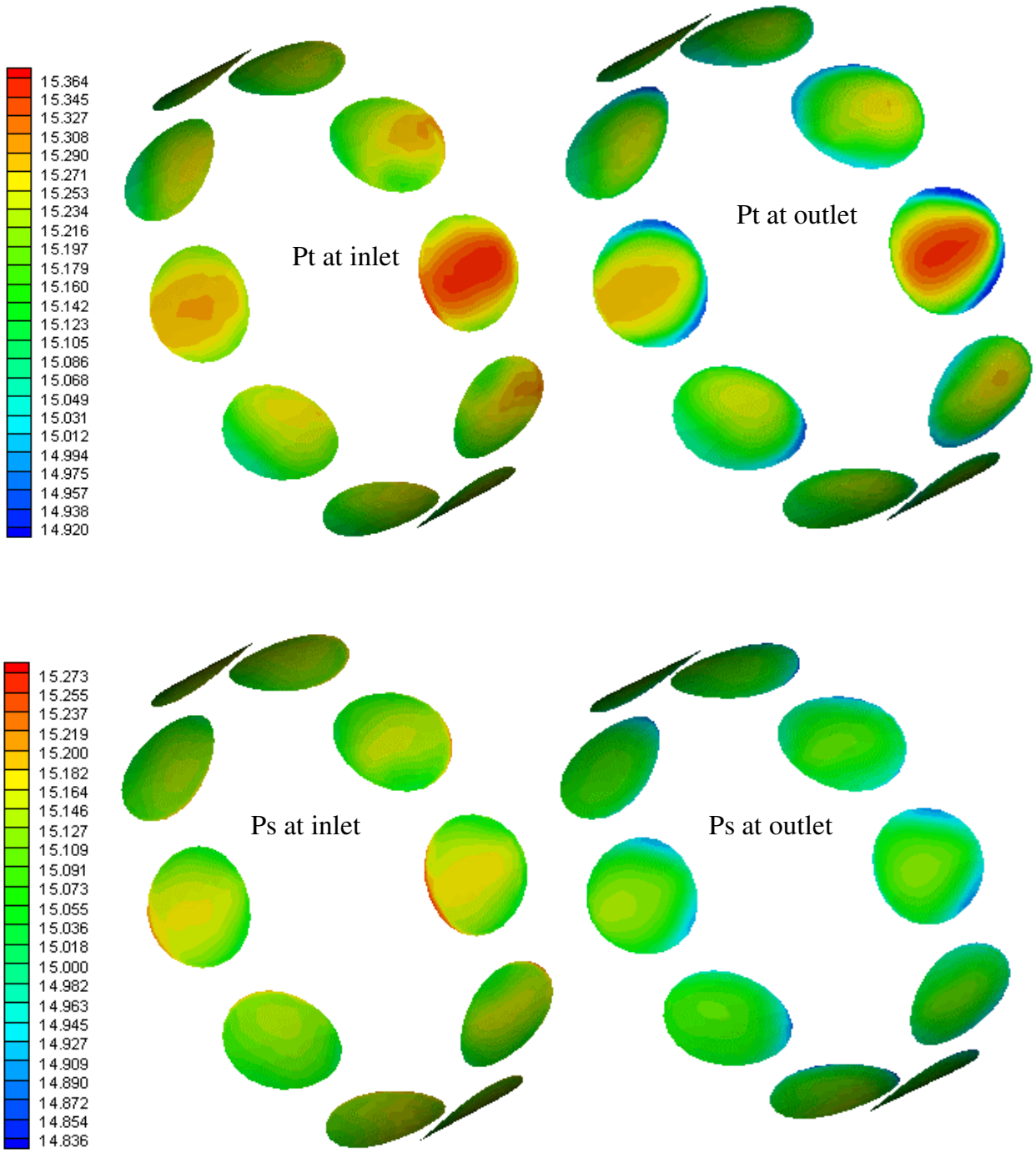


Figure 13: Typical contours of total and static pressure contours around the mixer holes – inlets and outlets.

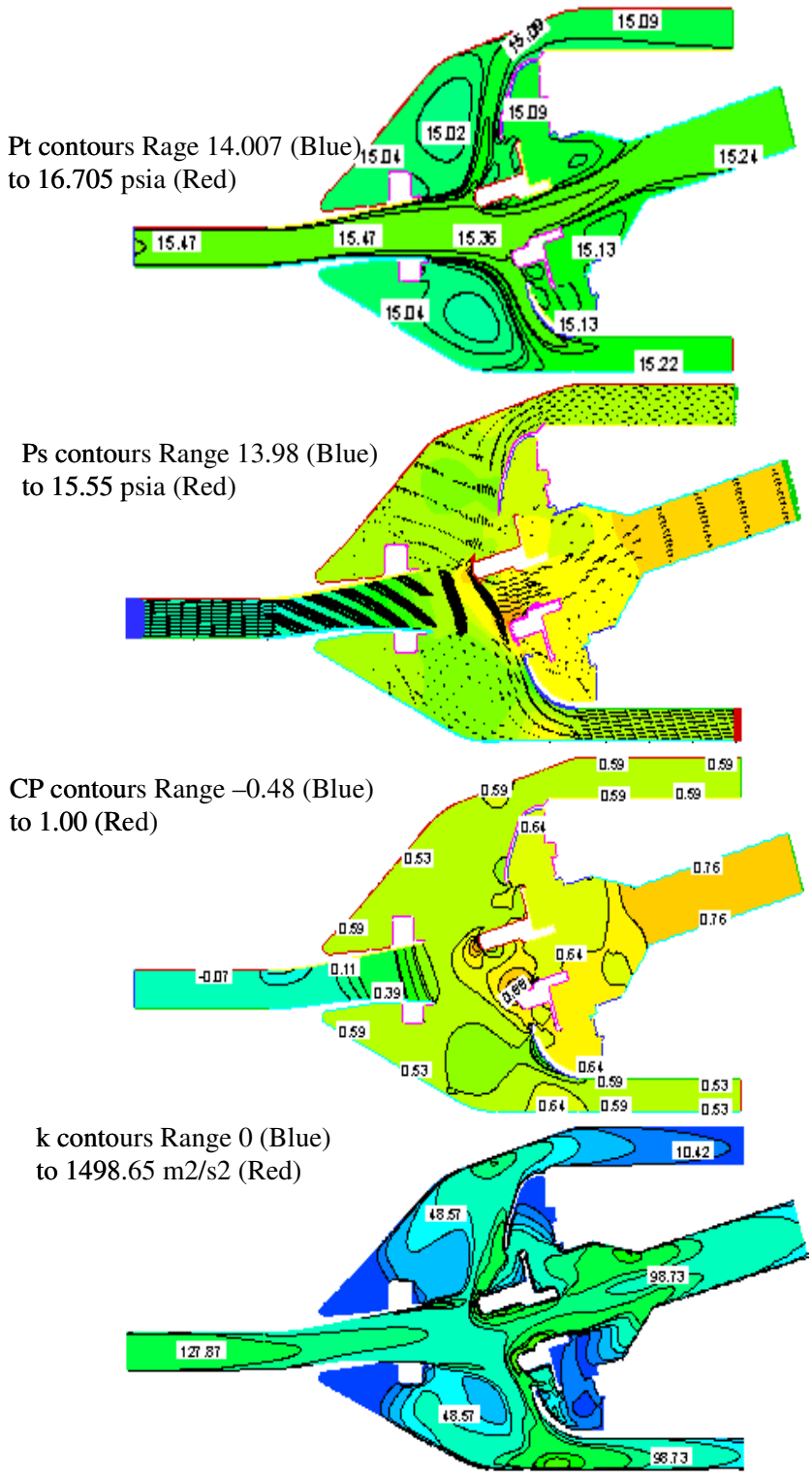


Figure 14: Typical contours of Pt, Ps (superimposed velocity vectors), CP and turbulence kinetic energy from 2d calculations.